

Detekt Static-Analysis Triage —

2026-06-04

First-ever Detekt run on the Lava Kotlin tree. Detekt is wired through the `StaticAnalysisConventionPlugin` buildSrc convention (applied by `lava.kotlin.library`, `lava.android.library`, `lava.android.application`, and transitively `lava.android.feature`), so **every module** is analysed with zero per-module configuration.

- **Plugin:** `io.gitlab.arturbosch.detekt 1.23.8` (Kotlin 2.1 compatible).
- **Config:** `config/detekt/detekt.yml` (detekt's curated default + `buildUponDefaultConfig = true`; one tuning — `FunctionNaming.ignoreAnnotated: ['Composable']` so `PascalCase @Composable` functions are not flagged, since Lava is 100% Jetpack Compose).
- **Baseline:** one `detekt-baseline.xml` **per module** (detekt's idiomatic multi-module pattern — a single shared baseline is impossible because each module's `detektBaseline` task *rewrites*, not merges, its target file). 45 baseline files, **674 deduplicated finding IDs** captured.
- **Severity:** detekt's defaults are uniformly warning; the gate fails on any finding via `build.maxIssues: 0`. The baseline suppresses today's findings so the gate is **green going forward** while any NEW finding fails the build.

Run commands

```
# Surface findings (first run, no baseline):
nice -n 19 ./gradlew --no-daemon --max-workers=2 --build-cache
detekt

# Capture baseline (serial not required for per-module files):
nice -n 19 ./gradlew --no-daemon --max-workers=2 --continue
detektBaseline

# Confirm green:
nice -n 19 ./gradlew --no-daemon --max-workers=2 --build-cache
detekt    # BUILD SUCCESSFUL
```

Findings by rule (raw console findings: 849)

The 849 raw console findings collapse to **674** baseline <ID>s because detekt deduplicates structurally-identical findings (same rule + signature) per module.

Rule	Count
MagicNumber	383
EmptyFunctionBlock	76

Rule	Count
LongParameterList	63
MaxLineLength	53
LongMethod	45
TooManyFunctions	42
TopLevelPropertyNameing	34
UnusedPrivateMember	33
TooGenericExceptionCaught	31
ReturnCount	24
CyclomaticComplexMethod	21
SwallowedException	9
UseCheckOrError	6
ThrowsCount	4
ComplexCondition	4
UnusedPrivateProperty	3
MatchingDeclarationName	3
ForbiddenComment	3
PrintStackTrace	2
NestedBlockDepth	2
FunctionOnlyReturningConstant	2
ThrowingExceptionsWithoutMessageOrCause	1
SpreadOperator	1
InvalidPackageDeclaration	1
ImplicitDefaultLocale	1
DeconstructingDeclarationWithTooManyEntries	1
ConstructorParameterNaming	1

What was FIXED vs BASELINED

Fixed (source code): 0. This change is scoped (by the worktree mandate) to buildSrc/, gradle/libs.versions.toml, and config/detekt/ — it MUST NOT touch .kt source files. Every finding is therefore **baselined**, and the genuine correctness/potential-bug findings are recorded below as backlog for a later source-touching phase (per the §6.D / §6.T.4 discipline: fixes land with a reproducing test, not in a build-wiring commit).

One **config-level** correction was applied (not a source fix): the Compose FunctionNaming false-positive class (10 findings on @Composable PascalCase functions in api-app) is suppressed by ignoreAnnotated: ['Composable'] rather than baselined — these are not findings at all, they are the Compose convention.

Baselined: all 674 deduplicated IDs. Breakdown by class:

- **Style / complexity (deferred, not bugs):** MagicNumber (383), MaxLineLength (53), LongMethod (45), LongParameterList (63), TooManyFunctions (42), TopLevelPropertyNameing (34), ReturnCount (24), CyclomaticComplexMethod (21), ThrowsCount (4), ComplexCondition (4), NestedBlockDepth (2), MatchingDeclarationName (3), FunctionOnlyReturningConstant (2), ForbiddenComment (3), SpreadOperator (1), InvalidPackageDeclaration (1), ConstructorParameterNaming (1), DestructuringDeclarationWithTooManyEntries (1).
- **Intentional empty bodies:** EmptyFunctionBlock (76) — the large majority are mDNS / NSD listener interface overrides (NsdMdnsAdvertiser) and test no-op callbacks where an empty body is the correct implementation. Removing them would break the override contract, so they are baselined, not "fixed".
- **Dead code (review needed):** UnusedPrivateMember (33), UnusedPrivateProperty (3) — removal is behavior-safe but must be reviewed per-site (some are Compose preview helpers / reflection targets); deferred.
- **Correctness / potential bugs (top backlog — see below):** TooGenericExceptionCaught (31), SwallowedException (9), UseCheckOrError (6), PrintStackTrace (2), ImplicitDefaultLocale (1), ThrowingExceptionsWithoutMessageOrCause (1).

Top backlog — correctness/potential-bug findings (fix in a later source phase)

These are the highest-value findings. Each **MUST** be fixed with a reproducing test per §6.T.1 / §6.D when a source-touching phase opens.

1. **ImplicitDefaultLocale — core/tracker/rutacker/.../domain/Utils.kt:102** (HIGH). `String.format("%.1f %sB", ...)` uses the implicit default locale to format file sizes. On a locale whose decimal separator is a comma (e.g. tr, de, ru — and Lava targets Russian-tracker users), `%.1f` emits `"1,5"` instead of `"1.5"`, corrupting displayed sizes and any downstream parse. Real user-visible latent bug. Fix: pass `Locale.US` (or `Locale.ROOT`).
2. **PrintStackTrace ×2 — RuTrackerNetworkApi.kt:87, RuTrackerAuth.kt:66** (MEDIUM). `Debug.printStackTrace()` calls in production network/auth paths. Per §6.AC these **MUST** route through the analytics non-fatal channel (`AnalyticsTracker.recordNonFatal`), not `stderr`. Behavior-changing; deferred to a §6.AC telemetry pass.
3. **SwallowedException ×9 + TooGenericExceptionCaught ×31** (MEDIUM). Broad catch (`e: Exception`) / swallowed exceptions across tracker + network code. Each site needs review against §6.AC (record non-fatal) — exactly the silent-fallback class §6.AC exists to surface. Highest-count correctness backlog.

4. **UseCheckOrError** ×6 (LOW). `throw IllegalStateException(...)` that should be `check()` / `error()` — idiomatic, behavior-equivalent, low-risk; e.g. `core/network/impl/SwitchingNetworkApi.kt:179,227`, `core/tracker/client/LavaTrackerSdk.kt:540`, `core/tracker/gutenberg/.../GutenbergSearch.kt:43` + `GutenbergBrowse.kt:66`.
5. **ThrowingExceptionsWithoutMessageOrCause** ×1 — `core/testing/.../TestAuthService.kt:11` (LOW, test fake). Default-constructor exception in a `core:testing` fake; per the Anti-Bluff Pact Third Law, fakes must mirror real behavior — add a message.

Maintenance notes

- To re-baseline after legitimately fixing findings: `./gradlew --no-daemon detektBaseline` then commit the shrunk `detekt-baseline.xml`.
- New findings in new code fail the build (not in baseline) — the gate is anti-bluff by construction: it only passes because today's findings are explicitly recorded, not because the rule set is empty.